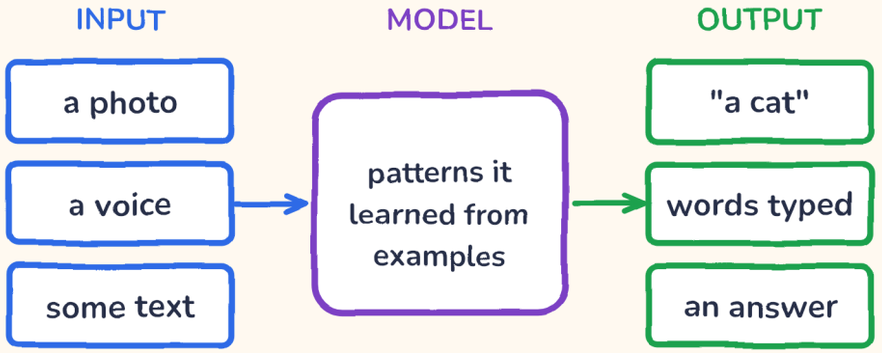
Artificial Intelligence · Machine Learning Foundations

# AI Basics

## How can a machine see, hear and decide? And what can't it do?

60 minutes | Pairs | Laptop + webcam




sense, think, act: the same loop as a robot

it always gives its best guess, never a certainty

Did you know?

The first program to learn by itself played draughts. Arthur Samuel wrote it in 1959, and it learned to beat him by playing thousands of games against itself.

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Let's think

# Is a calculator smart?

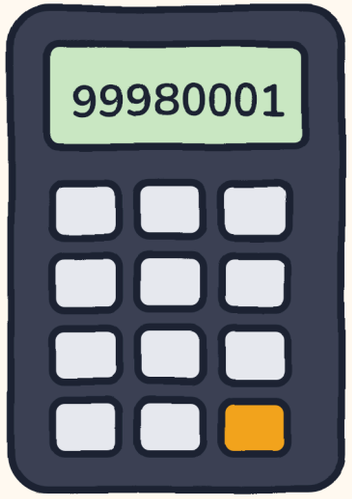
### Vote with your hands

Hands up if you think each one is **smart**:

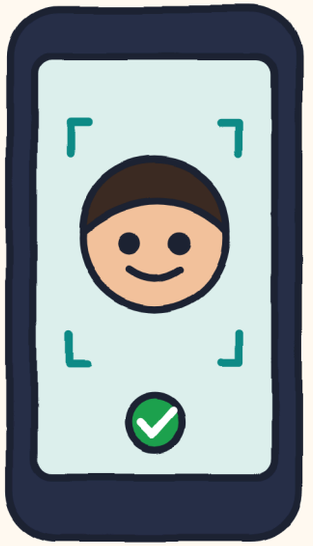
- a calculator that does 9,999 × 9,999 instantly
- a fan that turns on when it gets hot
- a phone that unlocks when it sees your face
- a parrot that says hello

### Think about it

- What makes a **person** smart?
- Is fast the same as smart?




very fast, but only follows rules



learned your face from examples

Hold on to your votes

By the end you will have a test for which of these are AI.

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Learning aims

# Today I will learn to...

1  Explain what **artificial intelligence** is, and what makes it different from a normal program

2 → Describe how AI works: **input, model, output**

3 ☆ Name **five things AI can do**, with an example of each

4  Say what AI **gets wrong**, and how to use it **safely**

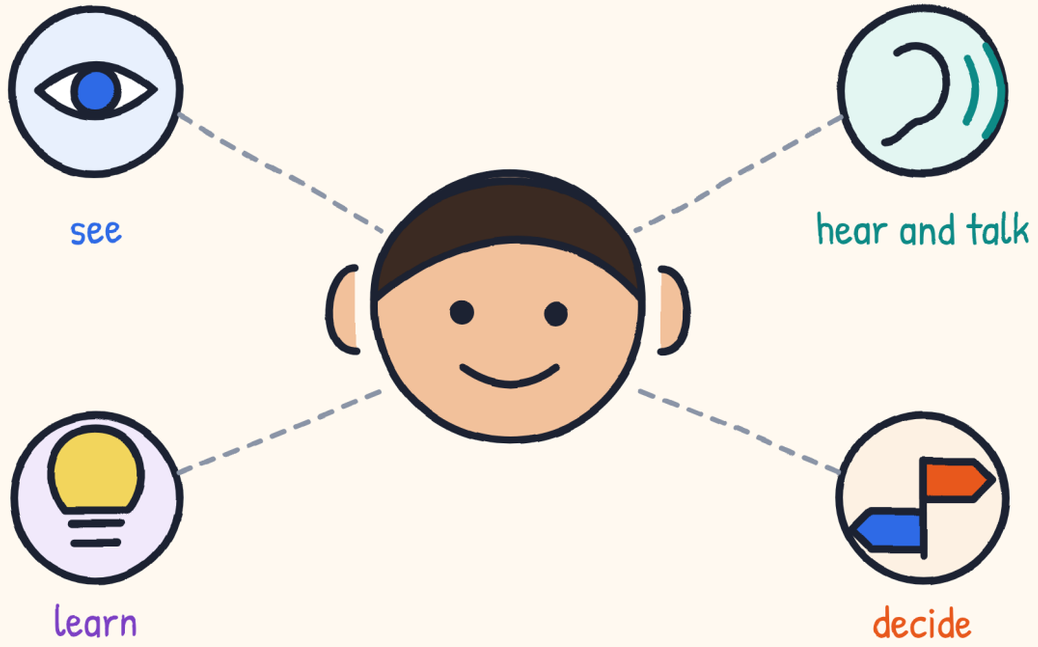
Keywords

intelligence · artificial intelligence · data · model · input · output · narrow AI · bias · prediction

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Objective 1 · The big idea

# What is intelligence?




## You do four things

- **See** a friend's face across the playground
- **Hear** what someone says, and answer
- **Learn** by practising and seeing examples
- **Decide**: cross the road now, or wait?

**Intelligence**

Taking in information, learning from it, and using it to decide what to do. You do it hundreds of times a day.

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Objective 1 · The big idea

# So what is AI?

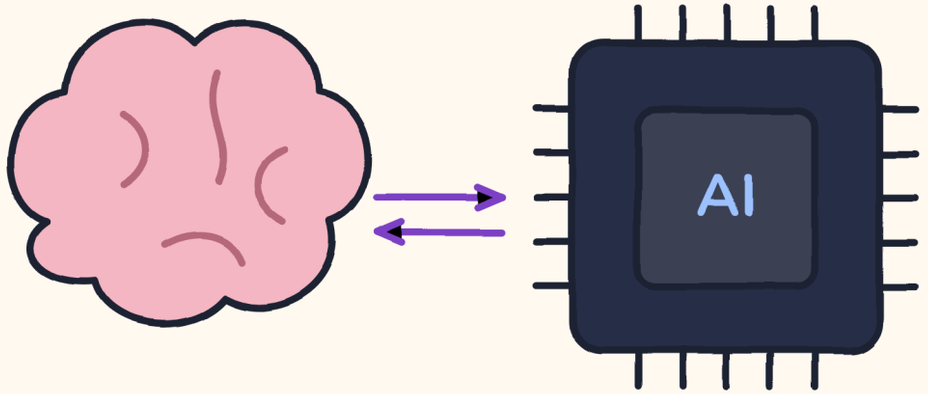
## Artificial intelligence

**Artificial** means made by people. **Intelligence** means seeing, hearing, learning and deciding.

So AI is **a computer program that does things we would call smart if a person did them**.

## AI is a tool

People build it, train it and decide how to use it. It has no feelings, no wishes and no plan of its own.




learns from experience

learns from data, made by people

The difference

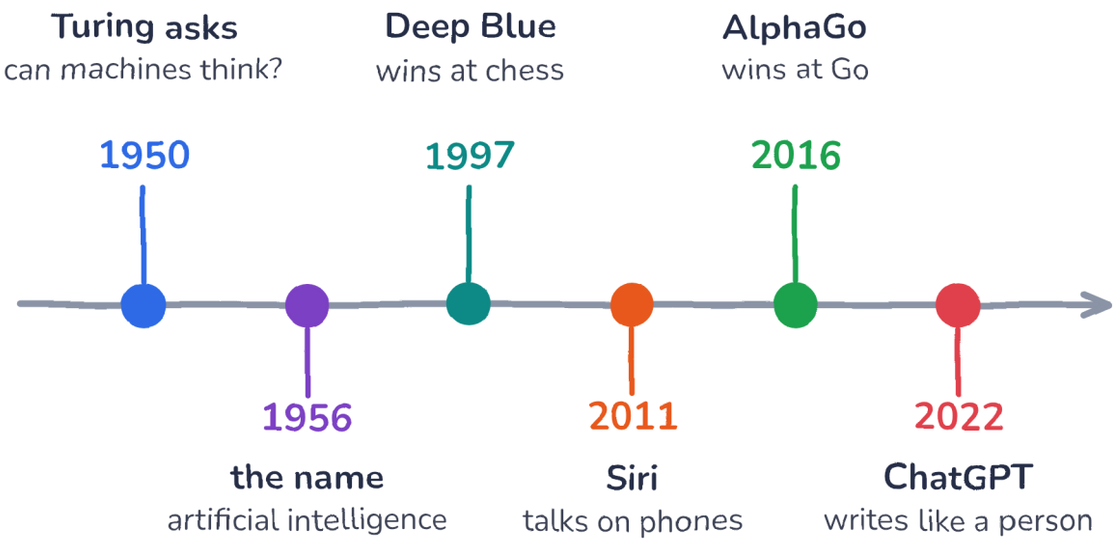
A normal program follows rules a person wrote. Most AI **learns its own rules from examples**, like you learned what a cat is.

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Objective 1 · History

# Seventy years of AI

## From a question to your pocket




seventy years from a question to your pocket

## Why so long?

The ideas are old. What was missing was **enough examples** to learn from and **fast enough computers**. The internet and smartphones gave us both.

## Why now?

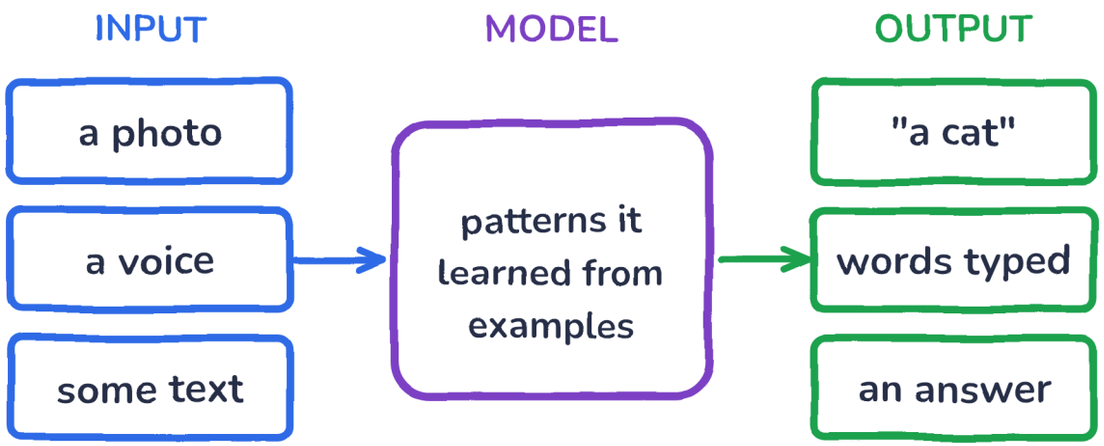
Every day people upload billions of photos and messages. A modern AI can learn from more examples in a week than a person sees in a lifetime.

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Objective 2 · How it works

# Input, model, output

## Every AI does these three things




sense, think, act: the same loop as a robot

it always gives its best guess, never a certainty

### 1. Input

What goes in: a photo from a camera, sound from a microphone, words you type.

### 2. Model

The patterns the AI learned during **training**, from thousands of examples.

### 3. Output

What comes out: a label, a sentence, an action. Always its **best guess**.

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Objective 3 · What AI can do

# Five things AI can do




see hear language decide create

**See**

Face unlock, photo search, a sick leaf on a farmer's phone.

**Hear**

Voice typing in Urdu, smart speakers, reading aloud.

**Language**

Google Translate, chatbots, spelling correction.

**Decide**

The next YouTube video, the fastest road, spam or not.

**Create**

Writing a story, drawing a picture, making music.

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Objective 3 · Kinds of AI

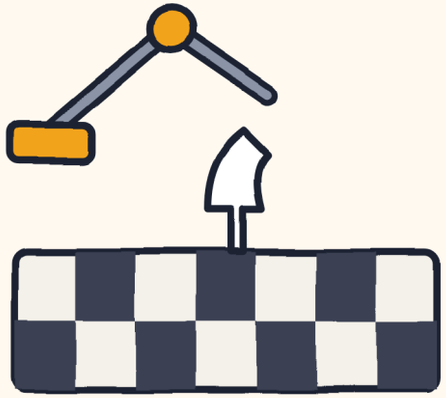
# Real AI versus film AI

### Narrow AI: every AI that exists

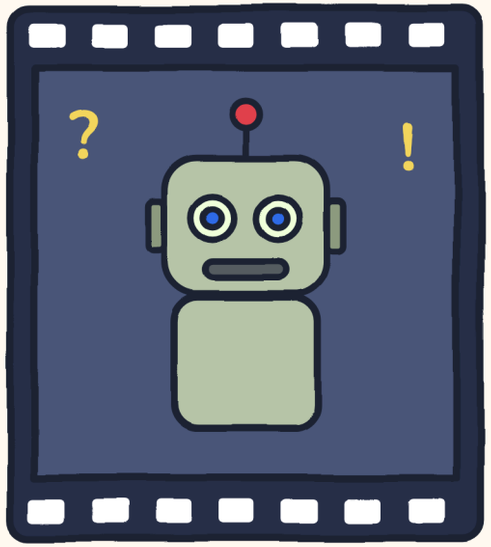
Very good at **one job**. A chess AI can't make tea. Face unlock can't translate Urdu.

### General AI: only in films

As clever as a person at **everything**, with its own goals. Nobody has built one.



narrow AI: brilliant at one job




film AI: does everything (made up)

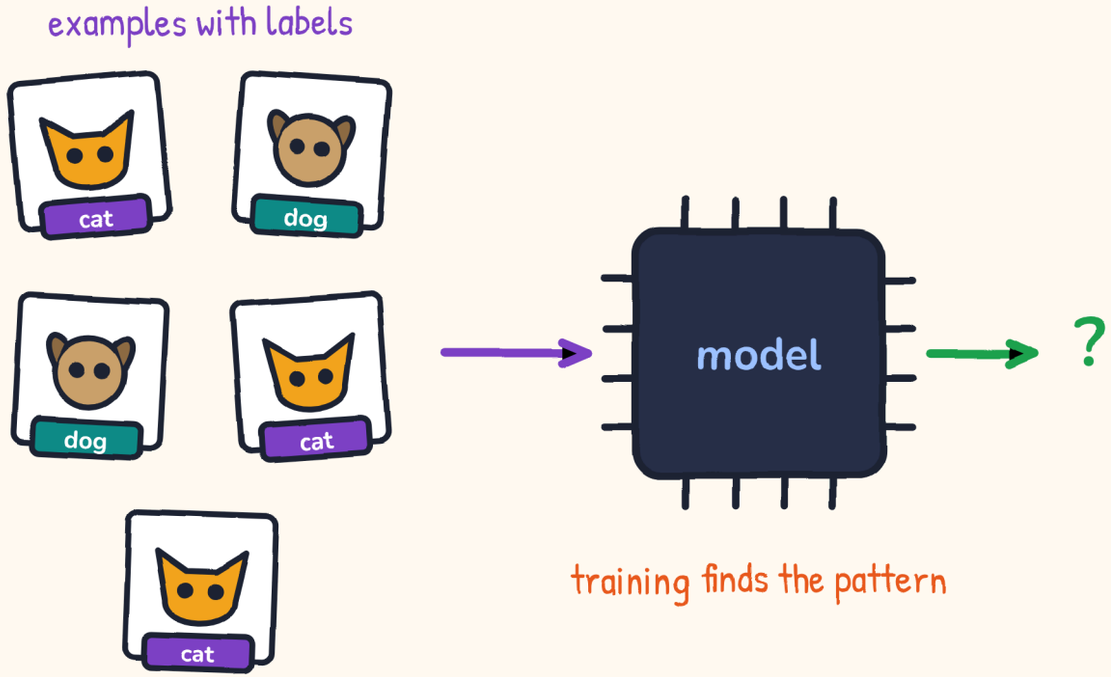
So when you hear "AI"

Ask: what one job was it trained for, and what examples did it learn from?

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Objective 2 · The fuel

# Data is what AI learns from




## Examples

Lots of them: photos, sounds, sentences. Thousands for a simple job, billions for a chatbot.

## Labels

The right answer for each example, written by a person.

Good data, good AI

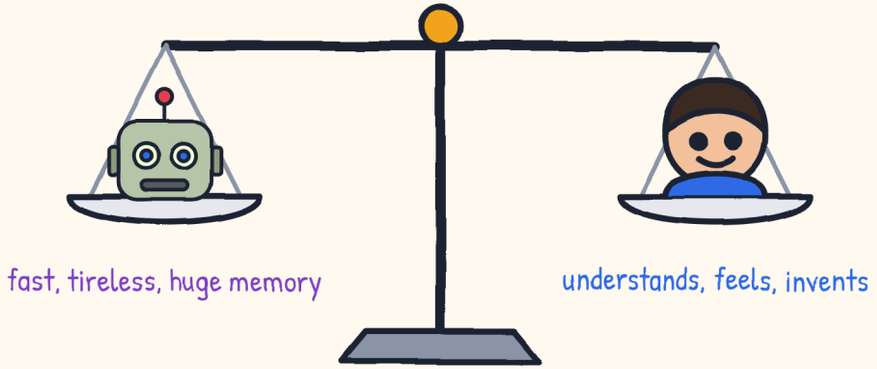
If the examples only show some people, the AI works well for them and badly for everyone else.

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Objective 4 · Humans and AI

# Who is better at what?

| Job | Better at it |
|---|---|
| Checking 10,000 photos for a fault | AI: fast, never tired |
| Spotting a pattern in a huge table | AI |
| Understanding a joke | Humans |
| Knowing a friend is upset | Humans |
| Doing a brand new kind of task | Humans |
| Deciding what is fair or kind | Humans: people stay in charge |



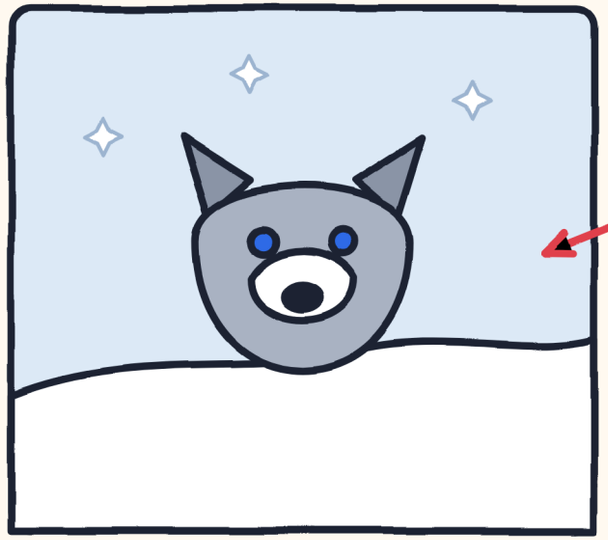

**The best results**

People and AI together: the AI does the fast part, a person checks and decides.

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Objective 4 · Limits

# What AI gets wrong



wolf
94% sure

it's a husky!

it learned: snow = wolf

every wolf photo it saw had snow

### Mistakes

It always gives its best guess, even when it is unsure.

### Bias

If the examples left some people out, it works worse for them.

### Made-up facts

Chatbots can write wrong answers that **sound** right.

**The habit to learn**

Treat an AI answer like advice from a stranger: useful, but check it before you trust it.

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Objective 4 · Using AI well

# Using AI safely and fairly

## Do

- **Check** facts in a book or a trusted website
- Say when you used AI
- Use it to explain what you're stuck on
- Ask an adult before trying a new AI app



keep private things private

## Don't

- Share your name, address or photos
- Hand in AI work as your own
- Believe it just because it sounds sure
- Make fake pictures of real people

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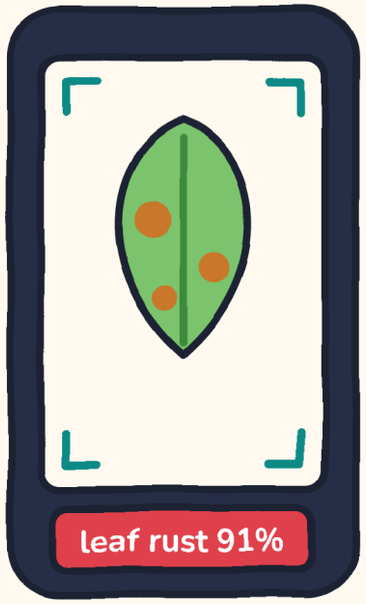
In the real world

# AI around you in Pakistan

snap a photo of a sick leaf

the app names the disease






## Farming

A farmer photographs a sick leaf, and the app names the disease so it can be treated early.

## And all around you

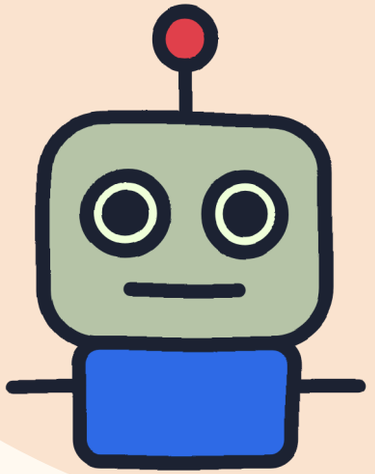
- **Travel:** Maps predicts traffic in Lahore and Karachi
- **Language:** voice typing and translation in Urdu
- **Shopping:** shops suggest what to buy next
- **Banking:** a strange payment gets flagged
- **Health:** AI helps read X-rays; a doctor checks

Part 2 · Make it

# Now let's meet AI

Decide who is better at what, sort AI by what it does, then teach a computer to see.

30 minutes | Pairs | Laptop + webcam



MAKE

Class game 1

# Human or AI: who is better?

## How to play

1. Left wall: **AI is better**. Right wall: **humans are better**. Middle: **both**.
2. The teacher reads a job. Walk to where you think it belongs.
3. Two people explain their choice. Anyone can change sides.
4. The teacher gives the answer, and why.

## Some jobs to read out

- Remembering every word in a dictionary
- Comforting a crying friend
- Finding one face in a crowd photo
- Inventing a new game

Notice

AI wins on speed and size. Humans win on understanding, feelings and anything new.

MAKE

Class game 2

# Sort the AI

## How to play

1. Each pair gets the app cards from the sheet.
2. Sort them into five piles: **see**, **hear and speak**, **language**, **decide**, **create**.
3. For each card, say what its **input** and **output** are.
4. Find one card that fits two piles, and explain why.

## Example

**Face unlock:** input a photo of a face, output *yes, it's you* or *no*. That's **see**.

Tricky one

A self-driving car fits every pile. Real machines often use several AIs together.

MAKE

Teach a computer to see

# Teachable Machine

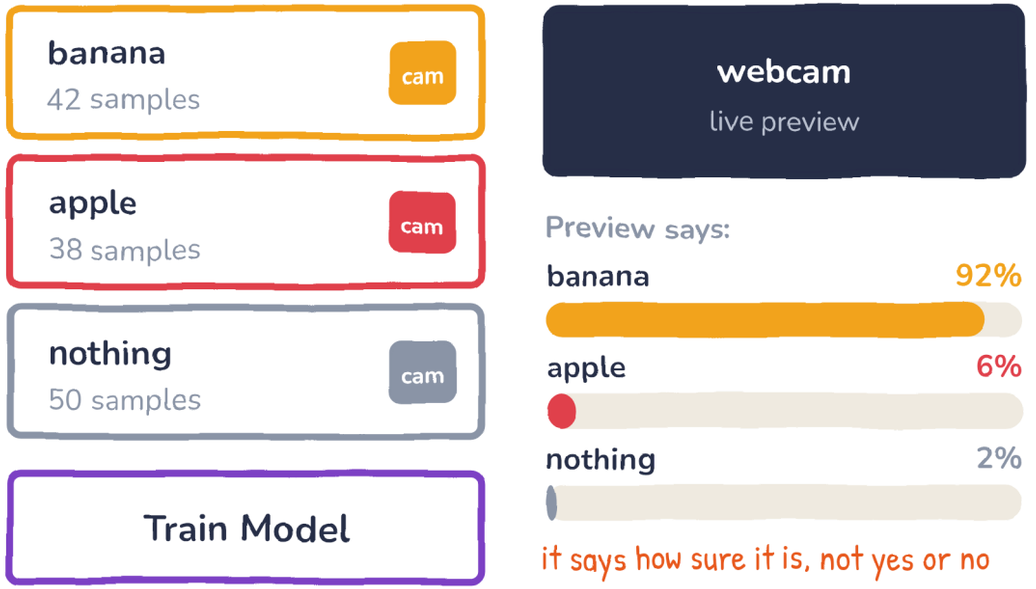
## Do this

1. Open **teachablemachine.withgoogle.com** → Image project → Standard.
2. Make two classes, like **thumbs up** and **thumbs down**.
3. Hold the webcam button: 30 pictures each, moving your hand.
4. Press **Train**, then test it. Try to trick it.

### Spot the three steps

Your pictures are the **input**. Training made the **model**. The bars are the **output**, with how sure it is.

## What you will see



LEVEL UP

Finished early?

# Go further

1 

Add a **third class** (an empty background) to Teachable Machine

Why does it need a 'nothing' class?

2 

Make it fail on purpose: train with **one person**, test with another

That is bias, made in two minutes.

3 

Find **three AIs** at home and write their input and output

TV, phone, car, kitchen: where else?

4 

Ask a chatbot a question you **know** the answer to

Was it right? How sure did it sound?

Check yourself

# Can I really do it? Prove it.

**1** Explain what **AI** is

Proof: I explained it to my partner in one sentence

**2** Describe **input, model, output**

Proof: I named all three for my Teachable Machine model

**3** Name **five things AI can do**

Proof: I sorted the app cards into the five piles

**4** Say what AI **gets wrong** and how to stay safe

Proof: I named one mistake and one safety rule

**Rate yourself:** red = not yet · amber = nearly · green = yes, and I can show someone

WRAP UP

Show what you know

# What did we learn?

## Five things to keep

- AI is a program that does things we would call smart, made by people.
- It works in three steps: input, model, output.
- It can see, hear, understand language, decide and create.
- All real AI is narrow: very good at one job.
- It learns from data, makes mistakes, and needs a person to check it.

## Exit ticket

Go back to the votes from the start. Which of the four was really **AI**, and why?

Next class

AI-03: Nearest neighbours. How a computer decides by comparing something new with things it has seen before.